Then deep inside the hole the region of space containing the star pinches off from our Universe and forms a small, closed universe with no connection to anything else (c). That closed universe then moves through hyperspace (d, e) and attaches itself to another large universe (I); and the star then explodes outward into that other universe (g, h).

I am uncomfortably aware that this scenario sounds like pure science fiction. However, just as black holes were a natural outgrowth of Schwarzschild's solution to the Einstein field equation (Chapter 3), so also this scenario is a natural outgrowth of another solution to the Einstein equation, a solution found in 1916–18 by Hans Reissner and Gunnar Nordström but not fully understood by them. In 1960 two of Wheeler's students, Dieter Brill and John Graves, deciphered the physical meaning of the Reissner–Nordström solution, and it soon became obvious that, with modest changes, the Reissner–Nordström solution would describe the imploding/exploding star of Figure 13.4. This star would differ from that of Oppenheimer and Snyder in just one fundamental way: It would contain within itself enough electric charge to produce a strong electric field when it gets highly compacted, and that electric field seemed in some way to be responsible for the star's reexplosion into another universe.

Let us take stock of where things stood in 1964, in the quest for Wheeler's holy grail—the quest to understand the ultimate fate of a star that implodes to form a black hole:

1. We knew one solution of Einstein's equation (the Oppenheimer–Snyder solution) which predicts that, if the star has a highly idealized form, including a perfectly spherical shape, then it will create a singularity with infinite tidal gravity at the hole's center—a singularity that captures, destroys, and swallows everything that falls into the hole.
2. We knew another solution of Einstein's equation (an extension of the Reissner–Nordström solution) which predicts that, if the star has a somewhat different highly idealized form, including a spherical shape and electric charge, then deep inside the black hole the star will pinch off from our Universe, attach itself to another universe (or to a distant region of our own Universe), and there reexplode.
3. It was far from clear which, if either, of these solutions was "stable against small, random perturbations" and thus was a candidate for